Machine Learning Fundamentals

Lecture 1: What is Machine Learning?

Machine learning (ML) is a field of artificial intelligence in which computers learn patterns from data instead of being explicitly programmed with rules. A model is trained on examples and then used to make predictions on new, unseen data.

There are three main types of machine learning. Supervised learning uses labeled data, where each example has an input and a known output. Unsupervised learning finds structure in unlabeled data, for example grouping customers into clusters. Reinforcement learning trains an agent to take actions in an environment so as to maximize a cumulative reward.

Typical supervised tasks are classification, where the output is a category such as spam or not spam, and regression, where the output is a continuous number such as a house price.

Lecture 2: Linear Regression

Linear regression models the relationship between input features and a continuous target as a straight line: y = w * x + b, where w is the weight (slope) and b is the bias (intercept).

The model is trained by minimizing the Mean Squared Error (MSE), which is the average of the squared differences between predicted and actual values.

Gradient descent is an optimization algorithm that updates the parameters in the direction of the negative gradient of the loss. The learning rate controls the size of each step: a learning rate that is too large can diverge, while one that is too small converges very slowly.

Lecture 3: Overfitting and Evaluation

Overfitting happens when a model learns the training data too well, including its noise, and performs poorly on new data. Underfitting happens when a model is too simple to capture the underlying pattern.

To detect overfitting we split the data into a training set, a validation set and a test set. A common split is 70% training, 15% validation and 15% test. K-fold cross-validation splits the data into k folds and trains k times, each time holding out a different fold.

Regularization reduces overfitting by penalizing large weights. L1 regularization (Lasso) can drive some weights to exactly zero, while L2 regularization (Ridge) shrinks all weights smoothly.

For classification, accuracy is the fraction of correct predictions. Precision is TP / (TP + FP), recall is TP / (TP + FN), and the F1 score is the harmonic mean of precision and recall.